

Java FSD - Day 7 Summary

Pattern Printing

- Pattern printing is one of the important topics in Java. It helps us understand how loops work together.
- In pattern programs, we generally use two loops: Outer Loop for rows and Inner Loop for columns.
- The print statement is usually written inside the inner loop.

Nested For Loop

- A nested for loop means writing one for loop inside another for loop.
- It is mainly used for matrix operations and pattern printing programs.
- For example, in a 3×3 matrix, the outer loop runs for rows and the inner loop runs for columns.

Flow of Execution

- First, the outer loop starts.
- Then, the inner loop runs completely.
- After the inner loop is completed, the outer loop moves to the next iteration.
- This process continues until the condition becomes false.

Reverse Pattern Printing

- Reverse pattern printing is used to display output in descending order.
- In this type of program, we usually use the decrement operator (--).
- Example: Printing numbers or stars from 5 to 1.

Important Points

- Nested loops are very useful for solving pattern-based questions.
- Pattern programs are frequently asked in interviews and coding assessments.
- Regular practice of star patterns, number patterns, and pyramid patterns will improve logical thinking and coding skills.

Note: Day 7 mainly focused on understanding the execution of nested loops and applying them in pattern printing programs.

Handwritten Notes

DAY - 7

2025
JANUARY

Monday 27

Pattern Printing

Outer Loop - For Row
Inner Loop - For Column

↳ Print statement

Nested For loop 3x3 Matrix

$i = 1$ & $i \leq 3$ so $i++ \rightarrow 2$
 $j = 1$ & $j \leq 3$ so $j++ \Rightarrow 2$

$j = 2$ & $j \leq 3$ so $j++ \rightarrow 3$

$j = 3$ & $j \leq 3$ so $j++ \Rightarrow 4$

$j = 4$ & $j \leq 3$ so goes to i

$i = 2$ & $i \leq 3$ so $i++ = 3$



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Reverse Printing

```

i = 5 ; 5 >= 1 ; i -- → 4
{
    j = 1 ; j(1) <= 5 ; j ++ → 2
    j = 2 ; 2 <= 5 ; j ++ → 3
    j = 3 ; 3 <= 5 ; j ++ → 4
    j = 4 ; 4 <= 5 ; j ++ → 5
    j = 5 ; 5 <= 5 ; j ++ → 6x
}
→ i = 4 ; 4 >= 1 ; i -- → 3
{
    j loop x)
}

```

```

i = 3 ; 3 >= 1 ; i -- → 2
{
    j loop x)
}

```

```

i = 2 ; 2 >= 1 ; i -- → 1
{
    j loop x)
}

```

```

i = 1 ; 1 >= 1 ; i -- → 0
{
    j loop x)
}

```

